



Sawbones Solid Rigid Polyurethane Foam 30 pcf Density

Categories: [Other Engineering Material](#); [Composite Core Material](#); [Polymer](#); [Thermoset](#); [Polyurethane, TS](#); [Thermoset Polyurethane Foam, Unreinforced](#)

Material Solid Rigid Polyurethane Foam

Notes:

Primary Use

Solid rigid polyurethane foam is primarily used as an alternative test medium for human cancellous bone. These products are not intended to replicate the mechanical properties of human bone, however, it does provide consistent and uniform material with properties in the range of human cancellous bone. Relevant mechanical properties for comparison to human cancellous bone may depend on the particular test method that is being developed.

The ASTM F-1839-08 "Standard Specification for Rigid Polyurethane Foam for Use as a Standard Material for Testing Orthopaedic Devices and Instruments" states that, "The uniformity and consistent properties of rigid polyurethane foam make it an ideal material for comparative testing of bones screws and other medical devices and instruments."

Description

Solid rigid polyurethane foam has a closed cell content ranging from 96.0 to 99.9%. Foam is available in a range of sizes and densities, from 0.08 to 0.80 grams per cubic centimeter (5 to 50 pounds per cubic foot).

Standard block size: 13cm X 18cm X 4cm

Standard sheet size: 13cm X 18cm X 3mm

Other Notes

Water absorption ranges from 0.301 to 0.0 kg/m²

Foam meets ASTM F-1839-08. Block dimensions are within ± 2 mm and sheet thickness is within ± 0.3 mm.

Foam density may vary $\pm 10\%$.

Properties parallel to direction of the rise of foam.

Information provided by Sawbones.

Vendors: No vendors are listed for this material. Please [click here](#) if you are a supplier and would like information on how to add your listing to this material.

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Physical Properties	Metric	English	Comments
Density	0.480 g/cc	0.0173 lb/in ³	
Mechanical Properties	Metric	English	Comments
Tensile Strength	12.0 MPa	1740 psi	
Tensile Modulus	0.592 GPa	85.9 ksi	
Compressive Yield Strength	18.0 MPa	2610 psi	
Compressive Modulus	0.445 GPa	64.5 ksi	
Poissons Ratio	0.30	0.30	
Shear Modulus	0.870 GPa	126 ksi	
Shear Strength	7.60 MPa	1100 psi	